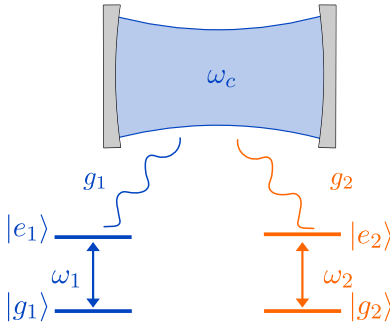


(a) Transverse coupling model



$$\hat{H} = \omega_c \hat{a}^\dagger \hat{a} + \sum_{j=1}^2 \frac{\omega_j}{2} \hat{\sigma}_z^{(j)} + \underbrace{\sum_{j=1}^2 g_j \hat{\sigma}_x^{(j)} (\hat{a} + \hat{a}^\dagger)}_{\hat{V}}$$

$$\Delta_j = \omega_j - \omega_c$$

$$\Sigma_j = \omega_j + \omega_c$$

Dispersive conditions

$$\frac{|g_j| \sqrt{n_* + 1}}{|\Delta_j|} \ll 1$$

$$\frac{|g_j| \sqrt{n_* + 1}}{\Sigma_j} \ll 1$$

(b) Second-order effective Hamiltonian

$$\hat{H}_{\text{eff}} = \hat{H}_0 + \frac{1}{2} [\hat{S}, \hat{V}] = \hat{H}_0 + \sum_{j=1}^2 \left[E_j^{(2)} + \frac{\chi_j}{2} \hat{\sigma}_z^{(j)} (\hat{a} + \hat{a}^\dagger)^2 \right] + J_{12}^{xx} \hat{\sigma}_x^{(1)} \hat{\sigma}_x^{(2)} + \mathcal{O}(g^3)$$

$$\hat{S} = \sum_{j=1}^2 \left[\frac{g_j}{\Delta_j} (\hat{a} \hat{\sigma}_+^{(j)} - \hat{a}^\dagger \hat{\sigma}_-^{(j)}) + \frac{g_j}{\Sigma_j} (\hat{a}^\dagger \hat{\sigma}_+^{(j)} - \hat{a} \hat{\sigma}_-^{(j)}) \right]$$

$$\chi_j = g_j^2 \left(\frac{1}{\Delta_j} + \frac{1}{\Sigma_j} \right)$$

$$J_{12}^{xx} = \frac{g_1 g_2}{2} \left(\frac{1}{\Delta_1} + \frac{1}{\Delta_2} - \frac{1}{\Sigma_1} - \frac{1}{\Sigma_2} \right)$$

Exchange sector

$$\hat{\sigma}_+^{(1)} \hat{\sigma}_-^{(2)} + \hat{\sigma}_-^{(1)} \hat{\sigma}_+^{(2)}$$

$$|e_1 g_2, n\rangle \leftrightarrow |g_1 e_2, n\rangle$$

Qubit-pair sector

$$\hat{\sigma}_+^{(1)} \hat{\sigma}_+^{(2)} + \hat{\sigma}_-^{(1)} \hat{\sigma}_-^{(2)}$$

$$\langle e_1 e_2, n | \hat{H}_{\text{eff}} | g_1 g_2, n \rangle \neq 0$$

Conditional cavity squeezing

$$(\hat{a} + \hat{a}^\dagger)^2$$

Normal sector $2\hat{a}^\dagger \hat{a} + 1$	 (squeezing) 	Anomalous sector $\hat{a}^2 + (\hat{a}^\dagger)^2$
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